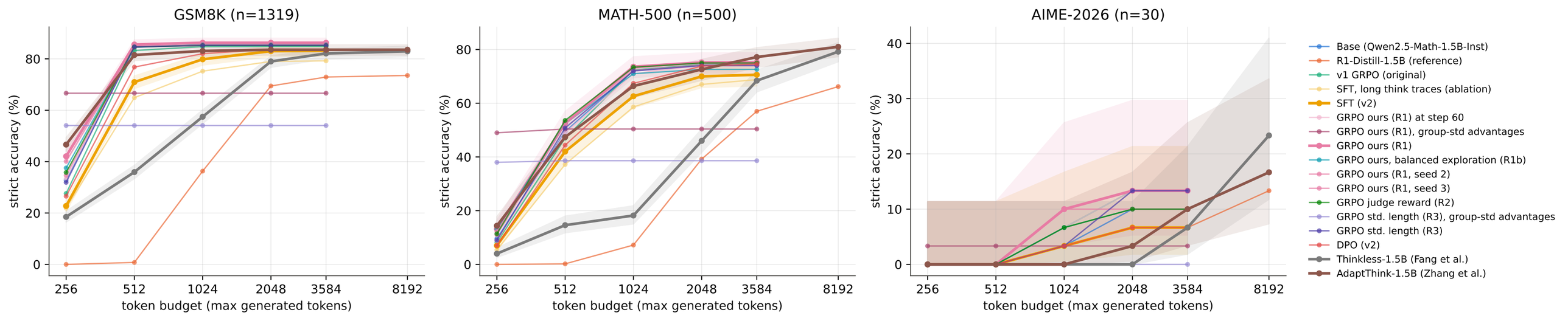
GSM8K (n=1319)
MATH-500 (n=500)
AIME-2026 (n=30)
strict accuracy (%)
token budget (max generated tokens)
256
512
1024
2048
3584
8192
0
20
40
60
80
10
30
Base (Qwen2.5-Math-1.5B-Inst)
R1-Distill-1.5B (reference)
v1 GRPO (original)
SFT, long think traces (ablation)
SFT (v2)
GRPO ours (R1) at step 60
GRPO ours (R1), group-std advantages
GRPO ours (R1)
GRPO ours, balanced exploration (R1b)
GRPO ours (R1, seed 2)
GRPO ours (R1, seed 3)
GRPO judge reward (R2)
GRPO std. length (R3), group-std advantages
GRPO std. length (R3)
DPO (v2)
Thinkless-1.5B (Fang et al.)
AdaptThink-1.5B (Zhang et al.)